ConsentChain: PureChain-Powered Verifiable Machine Unlearning and Dynamic Consent for Federated IoT Learning

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***Abstract*—Federated learning over consumer IoT devices is increasingly subject to regulatory regimes (GDPR Article 17, EU AI Act Article 15, CCPA § 1798.105) that require data subjects to withdraw consent and have the trained model demonstrably forget their contribution. Existing work treats consent management, machine unlearning, and on-chain auditability as three disjoint research threads. We close the *consent–unlearning–verification triangle* with ConsentChain, a protocol that binds the three layers in a single chain-enforced pipeline. Our contributions are: (i) a Consent Lifecycle State Machine (CLSM) modelling consent as a 6-state finite automaton implemented as soulbound ERC-721 tokens; (ii) a Verifiable Unlearning Protocol (VUP) that gates on-chain acceptance on three independent membership inference attacks; (iii) a three-contract Solidity architecture deployed on PureChain (gas-free, PoA^2) and Polygon Amoy (gas-based, PoS) for the first apples-to-apples cost comparison of consent-aware federated learning under both economic models; and (iv) an end-to-end dual-domain empirical evaluation on a 9-client N-BaIoT autoencoder federation (primary) cross-validated on a 5-client CASAS BiLSTM activity classifier, across three approximate-unlearning paradigms plus an exact-retraining baseline. We observe per-data-subject lifecycle costs of $\$0$ on PureChain versus \$0.0012 on Polygon Amoy and per-round on-chain validation costs that scale linearly with the participant count N, reaching \$0.0176 at $N = 500$. The protocol accepts approximate-unlearning attestations (MIA < 0.55) while rejecting both naive retraining (whose distributional uniqueness raises false-positive MIA scores) and any attestation lacking a 3-of-5 verifier threshold, demonstrating that the on-chain gate is faithful to MIA evidence rather than to the data controller's claim.**

***Index Terms*—federated learning, machine unlearning, smart contracts, dynamic consent, IoT, GDPR, EU AI Act, membership inference attack**

I. Introduction

Federated learning (FL) trains a shared model across decentralised clients without centralising their data [1]. In the consumer-IoT setting — home assistants, smart-home sensors, wearable biometrics, connected vehicles — this is existential: participants are unwilling, and increasingly not legally permitted, to upload raw data. The FL literature has accordingly focused on *technical* privacy (secure aggregation, leakage bounds, poisoning robustness) while a second, equally binding axis remains under-addressed: *regulatory* privacy. GDPR Article 7 requires consent to be demonstrably given, purpose-granular, and freely withdrawable; Article 17 grants erasure — which, applied to a trained model, means the model must *forget* a revoked participant's contribution. The EU AI Act [2] (Arts. 15/26) and CCPA [3] § 1798.105 impose analogous, auditable obligations. None of these regimes assumes the deployment *chooses* to comply; each demands demonstrable, audit-grade evidence.

Three research threads each address one corner of this problem but not the whole. Consent-management ledgers (MedRec [4]) record consent immutably but cannot gate downstream training. Machine-unlearning methods (SISA [5], FedEraser [6]) remove a participant's contribution but leave the trigger and verification off-chain. Blockchain-FL systems [7], [8] log model updates on-chain without consent semantics or forgetting verification. No published work binds the three.

We frame the gap as a missing *edge*, not a missing vertex. Consent must stay authoritative over which data enters which round; unlearning (by any method) must be *verifiable* rather than self-asserted; and verification — against membership inference, the canonical operational test for forgetting [9]–[11] — must in turn drive the consent state, closing the loop. Prior work implements the vertices; ConsentChain supplies the edges. Enforcing them off-chain would merely move trust to the data controller; putting the loop on a public smart-contract substrate removes that trust but introduces a cost question — a single revocation on Ethereum mainnet costs dollars, prohibitive at IoT scale. We show the identical protocol on a public, gas-free, EVM-compatible chain with PoA^2 consensus costs *exactly* \$0 per operation. This work make the following key contributions:

1) A **Consent Lifecycle State Machine** (CLSM): six states, nine transitions, four formally-stated safety and liveness properties, instantiated isomorphically as a soulbound ERC-721 contract.
2) A **Verifiable Unlearning Protocol** (VUP) that accepts an Unlearning Attestation iff three *independent* membership-inference attacks (shadow [9], LiRA [10], label-only [11]) all fall below τ, certified by t-of-n ECDSA signatures. It is method-agnostic over the unlearning algorithm.
3) A **three-contract Solidity architecture** (ConsentRegistry, UnlearningOracle, FLGatekeeper) deployed on *both* a gas-free chain (PureChain, PoA^2) and a gas-based chain (Polygon Amoy, PoS) — the first side-by-side cost comparison under both economic models.
4) A **dual-domain empirical evaluation**: a 9-client N-BaIoT [12] autoencoder federation (primary), cross-validated on a 5-client CASAS smart-home BiLSTM

classifier, over three approximate-unlearning paradigms plus an exact-retraining baseline — measuring per-method MIA forgetting, on-chain gas/USD cost on both chains, per-round validation cost to $N = 500$, and post-revocation utility recovery.

II. Related Work

Three converging research threads form the prior art for ConsentChain. We summarise each, then state the gap.

Consent management on distributed ledgers. MedRec [4] pioneered the use of a blockchain substrate to record patient consent for medical-data access and permissioning. The system established the durability and auditability properties we inherit: every consent state transition is an immutable log entry, and the consent owner retains exclusive control over modifications. MedRec, however, treats the consent record as the *endpoint* of the protocol — there is no downstream ML training whose execution is gated by the consent state, and no mechanism for demonstrating that withdrawn consent translated into a model update. Subsequent semantic-web work (GConsent, W3C DPV) formalises consent vocabularies but operates in an off-chain registry without enforcement. *Gap*: consent is recorded but does not influence training.

Machine unlearning. The SISA training scheme [5] pre-shards training data and caches per-shard intermediate models so that data removal requires retraining only the affected shard. FedEraser [6] extends SISA's logic to the federated setting by replaying aggregation history with a calibration step, significantly cheaper than full retraining. Gradient-ascent unlearning methods take a single trained model and perform a small number of negative-learning-rate steps on the data to be removed. All three families have been validated on standalone benchmarks but *none specifies how unlearning is triggered, by whom, or how a deployer demonstrates it actually happened*. The unlearning literature treats the trigger and verification steps as out-of-scope. *Gap*: unlearning is implementable but not verifiable end-to-end.

Blockchain-based federated learning. BlockFL [7] records every device-side model update on-chain to defend against malicious aggregators. BFLC [8] replaces the FL aggregator with a committee-consensus mechanism. These systems address aggregator-side trust but do not encode consent at all — a participant who joins the federation cannot subsequently withdraw without trusting the orchestrator to respect the request. Membership inference attacks [9]–[11] provide the canonical operational test for whether a model still memorises a specific data subject's training contribution, but no published blockchain-FL system uses MIA scores as the on-chain acceptance criterion for a state transition. *Gap*: model updates are auditable but consent semantics and forgetting verification are absent.

Positioning. ConsentChain is the first system, to our knowledge, that binds all three threads. We adopt MedRec's on-chain consent registry pattern (vertex A), extend it with the CLSM safety and liveness properties; we accept any of the three established unlearning paradigms as a black-box (vertex B); and we make the verification step (vertex C) the chain's gating condition for a consent state transition. The pattern that emerges — consent change → unlearning execution → three-MIA verification → on-chain attestation → consent state finalisation — is the missing edge.

The choice of *which* blockchain matters at IoT scale. Permissionless gas-based chains (Ethereum, Polygon) impose recurring per-operation costs that scale linearly with device count and consent operation rate. Section V quantifies this against PureChain [13], a public, gas-free, EVM-compatible chain with PoA^2 consensus and Merkle-proof-backed state. PureChain is not the contribution of this paper; rather, our contribution is showing that the protocol is operationally identical on both chain types, while the economic model is fundamentally different.

III. System Model and Threat Model

A. Entities

ConsentChain has six entities. The data subject DS_i controls the private key of one soulbound Consent NFT. The data controller DC orchestrates the FL round loop and triggers unlearning on revocation events. Each FL participant device FL_k runs local SGD and submits per-round updates. The Verifier Committee $VC = \{v_1, \ldots, v_n\}$ holds $n = 5$ registered ECDSA keys; threshold $t = 3$ by default. Three on-chain contracts — `ConsentRegistry` (CR), `UnlearningOracle` (UO), and `FLGatekeeper` (GK) — collectively implement the protocol. Anyone with chain read access is an *auditor*; public auditability is a first-class property.

B. Federated learning baseline

We adopt FedAvg [1] as the canonical aggregation rule. Let $\theta^{(r)}$ denote the global parameters at round r, $K_r \subseteq \{1, \ldots, K\}$ the set of validated participants returned by GK.`validateRound`, and n_k the local training-set size of client k:

$$\theta^{(r+1)} = \sum_{k \in K_r} \frac{n_k}{\sum_{j \in K_r} n_j} \theta_k^{(r+1)} \tag{1}$$

The novel restriction is that K_r must be the `validateRound` return value rather than an orchestrator-chosen subset — consent is enforced at the K_r membership level, not at the gradient level.

C. Consent Lifecycle State Machine

Definition 1 (CLSM)**.** *The Consent Lifecycle State Machine is the deterministic finite automaton* $\text{CLSM} = (S, \Sigma, \delta, s_0, F)$ *where:*

$$\begin{aligned} S &= \{\textsc{Inactive}, \textsc{Active}, \textsc{Suspended}, \\ &\qquad \textsc{Revoked}, \textsc{Expired}, \textsc{Unlearned}\}, \\ \Sigma &= \{\texttt{grant}, \texttt{modify}, \texttt{suspend}, \texttt{resume}, \texttt{revoke}, \\ &\qquad \texttt{tick}, \texttt{expire}, \texttt{unlearn_verified}\}, \\ s_0 &= \textsc{Inactive}, \qquad F = \{\textsc{Unlearned}\}. \end{aligned}$$

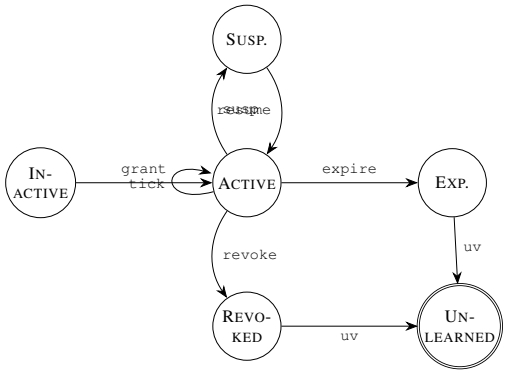


Fig. 1. Consent Lifecycle State Machine. `modify` shares the ACTIVE self-edge with `tick`; `uv` = `unlearn_verified` is callable only by the oracle. UNLEARNED (double circle) is absorbing.

TABLE I
THREAT MODEL SUMMARY (FULL VERSION IN SUPPLEMENTARY).

Adversary	Defense
Lazy FL Orchestrator	GK reverts on pending unlearning; $\Delta_{\max}$ timeout emits ComplianceViolation
Deceptive Controller	Three MIAs + VC t-of-n signatures
Colluding participants	Out of scope (orthogonal DP layer)
Consent forger	Soulbound NFT + `dsOwner` check
Replay attacker	UA binds (chainid, tokenId, hashPre)

$\delta : S \times \Sigma \to S$ is a partial function whose transition diagram appears in Figure 1 and whose tabular form is Table II in Section IV-A; UNLEARNED *is the absorbing terminal state.*

The on-chain contract `ConsentRegistry` is an exact instantiation of CLSM: each consent NFT carries a `ConsentState` enum stored in slot 0 of the `ConsentRecord` struct, every δ edge corresponds to one external function, and the soulbound constraint (no `transferFrom`) enforces that only DS_i may invoke transitions on token i.

D. Threat model

We adopt the adversary set summarised in Table I. Five in-scope adversaries are defended against by specific protocol mechanisms; six classes of attack are acknowledged as out-of-scope with the rationale stated in Section VI. The expanded model with per-adversary capability/intent/defense/residual-risk lives in the supplementary `docs/threat-model.md`.

E. Formal properties

We state four properties that ConsentChain must satisfy. Each is proved informally in Section IV; full mechanised proofs are future work.

Property 1 (Consent Integrity — safety)**.** *For every block height b and round r that starts at b via GK.`validateRound`,*

$$\forall i \in K_r : \quad CR.\texttt{state}(i) = \textsc{Active}_b \ \wedge \ \rho_r \in \mathcal{P}_i, \qquad (2)$$

where ρ_r is the round purpose passed to `validateRound` and $\mathcal{P}_i$ is the purpose set of the consent token i recorded at `grantConsent` time. No transition path in δ leads from REVOKED *back to* ACTIVE*.*

Property 2 (Unlearning Enforceability — liveness)**.** *For every revocation event recorded at block b on token i,*

$$\exists\, b' \le b + \Delta_{\max} : CR.\texttt{state}(i) = \textsc{Unlearned}_{b'} \\ \vee \ \ GK.\texttt{ComplianceViolation}(i)_{b'}. \qquad (3)$$

Equivalently: either a valid UA is committed within $\Delta_{\max}$ blocks of the revocation, or the chain logs a publicly verifiable ComplianceViolation event that any auditor can pick up. The system degrades to "no training" rather than "unconsented training."

Property 3 (Unlearning Verifiability)**.** *The `UnlearningOracle` accepts a UA submission (u, Σ) if and only if*

$$\big(\forall j \in \{\mathrm{sh}, \mathrm{lira}, \mathrm{lo}\} : \ s_j < \tau\big) \wedge \big|\{\sigma \in \Sigma : \mathrm{verify}(\sigma, h_u, VC)\}\big| \ge t, \qquad (4)$$

where s_j are the three MIA scores in the UA, h_u is the UA hash (Eq. 10), and verifications are over the registered verifier set VC of size n.

Property 4 (Audit Completeness)**.** *Every consent state transition, every Unlearning Attestation acceptance, every ComplianceViolation, and every per-round model hash update emits a Solidity event indexed by token ID and round ID.*

Properties 1 and 2 are the safety and liveness halves of the classical agreement-protocol decomposition. Property 3 is the cryptographic gating condition. Property 4 ensures that the chain serves as the durable evidence layer required by GDPR Article 5(2) accountability principle and EU AI Act Article 12 record-keeping. The empirical evidence for 2 and 3 is reported in Section V; the on-chain enforcement of 1 and 4 follows by construction from the contract code (Section IV).

IV. THE CONSENTCHAIN FRAMEWORK

We now detail the three layers of ConsentChain: the CLSM and its on-chain instantiation (Section IV-A), the Verifiable Unlearning Protocol with its three MIA components (Section IV-B), and the three-contract Solidity architecture that closes the automation loop (Section IV-C). Figure 2 gives the layered overview.

A. Consent Lifecycle State Machine

The CLSM transition function δ from Definition 1 is summarised in Table II. Five edges are DS_i-controlled (`grant`, `modify`, `suspend`, `resume`, `revoke`); two are time-driven (`tick`, `expire`); one (`unlearn_verified`) is callable only by the `UnlearningOracle` after a valid UA submission. The UNLEARNED state has no outgoing edges, making it absorbing.

Property 1 follows by inspection of Table II: there is no edge into ACTIVE from REVOKED or EXPIRED, and the `tick`(r) self-edge guards on $\rho_r \in \mathcal{P}$. The `validateRound` function (Section IV-C) fires only the `tick` edge.

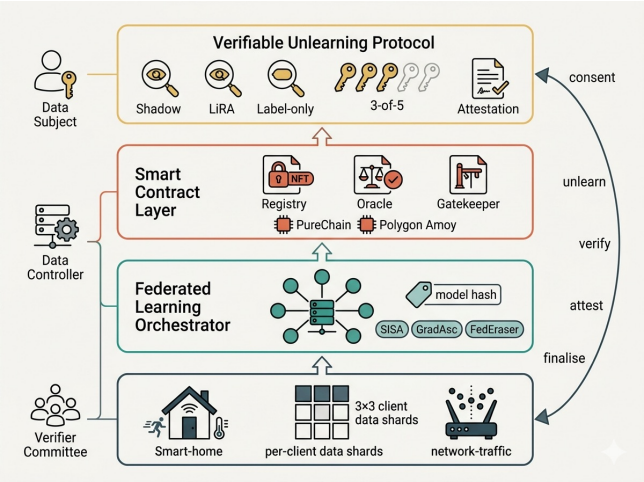


Fig. 2. ConsentChain architecture: off-chain IoT data and Flower FedAvg layers, the three on-chain Solidity contracts on PureChain / Polygon Amoy, and the off-chain Verifiable Unlearning Protocol. The clockwise loop consent→unlearn→verify →attest→finalise is the edge ConsentChain closes.

TABLE II
CLSM TRANSITION TABLE. REV = REVOKE, SUSP = SUSPEND, UV = UNLEARN_VERIFIED.

Current	Event	Next	Guard
INACTIVE	`grant`$(\mathcal{P}, T)$	ACTIVE$(\mathcal{P}, T)$	DS signature
ACTIVE	`modify`$(\mathcal{P}', T')$	ACTIVE$(\mathcal{P}', T')$	DS signature
ACTIVE	`susp()`	SUSPENDED	DS signature
ACTIVE	`tick`(r)	ACTIVE	$\rho_r \in \mathcal{P}$, $T > b$
ACTIVE	`rev`$(\mathcal{P}')$	REVOKED$(\mathcal{P}')$	DS signature
ACTIVE	`expire()`	EXPIRED	$b > T$
SUSPENDED	`resume()`	ACTIVE	DS signature
REVOKED	`uv()`	UNLEARNED	valid UA on-chain
EXPIRED	`uv()`	UNLEARNED	valid UA on-chain

B. Verifiable Unlearning Protocol

When a token transitions to REVOKED or EXPIRED, the gatekeeper records `pendingUnlearningSince`$[i] = b$ and `validateRound` blocks all future rounds (Property 2). The data controller has until $b + \Delta_{\max}$ to (i) execute an unlearning procedure, (ii) collect three MIA scores from the post-unlearning model, (iii) assemble the Unlearning Attestation, (iv) gather t-of-n verifier signatures, and (v) submit it on-chain.

1) Approximate unlearning methods: ConsentChain is method-agnostic. We evaluate three established families:

a) SISA-Fed: pre-shards each client's data into S disjoint shards and caches a per-shard intermediate checkpoint [5]. Unlearning a revoked client DS_i reduces to retraining only the shards that touched i's data, then re-aggregating.

b) Gradient ascent: performs k steps of negative-learning-rate gradient ascent on the cached gradients of the revoked client:

$$\theta^{(j+1)} = \theta^{(j)} + \eta \nabla \mathcal{L}\Big(\theta^{(j)}; \mathcal{D}_i\Big), \quad j = 0, 1, \ldots, k-1, \tag{5}$$

where $\mathcal{D}_i$ is DS_i's local training set, $\mathcal{L}$ is the FL loss (here mean-squared reconstruction error for the N-BaIoT autoencoder), and $\eta > 0$ is the unlearning rate. We use $k = 10$, $\eta = 10^{-4}$.

c) FedEraser: [6] replays the aggregation history excluding DS_i's contributions, then applies a short calibration step. We use 3 calibration rounds.

2) Three independent membership inference attacks: We evaluate the post-unlearning model f_{post} against DS_i's training set $\mathcal{D}_i$ (the "member set" M) using three attacks designed to be statistically independent.

a) Shadow-model MIA [9] (autoencoder generalisation [14]).: Train $S = 5$ shadow autoencoders on auxiliary disjoint subsets of the non-revoked clients' data; for each shadow, compute per-feature reconstruction errors on "in" and "out" subsets and use these as features for an MLP attack classifier. The attack score is the fraction of M that the classifier flags as "in:"

$$s_{\text{sh}} = \frac{1}{|M|} \sum_{x \in M} \mathbb{I}\big[A_{\text{sh}}\big(\phi(f_{\text{post}}, x)\big) = 1\big], \tag{6}$$

where ϕ extracts a 6-dimensional feature vector (per-feature MSE summary statistics) from the model's reconstruction.

b) LiRA [10].: Train $n_{\text{in}} = n_{\text{out}} = 4$ reference autoencoders. For each sample $x \in M$ fit per-example Gaussians $\mathcal{N}(\mu_x^{\text{in}}, \sigma_x^{\text{in}2})$ and $\mathcal{N}(\mu_x^{\text{out}}, \sigma_x^{\text{out}2})$ to the log-reconstruction-errors observed under in-pool and out-pool reference models. The log-likelihood ratio is

$$\Lambda(x) = \log p\big(\ell_{\text{post}}(x) \,\big|\, \mu_x^{\text{in}}, \sigma_x^{\text{in}}\big) - \log p\big(\ell_{\text{post}}(x) \,\big|\, \mu_x^{\text{out}}, \sigma_x^{\text{out}}\big), \tag{7}$$

where $\ell_{\text{post}}(x) = \log(\epsilon + \|f_{\text{post}}(x) - x\|^2)$. The threshold $\hat{\Lambda}_\alpha$ is the $(1-\alpha)$-quantile of Λ on the auxiliary set, giving an α-controlled FPR; the score is the empirical TPR at that FPR: $s_{\text{lira}} = |M|^{-1} \sum_{x \in M} \mathbb{I}[\Lambda(x) > \hat{\Lambda}_\alpha]$. We report $\alpha = 0.05$ as canonical and $\alpha = 0.5$ as the § VI ablation.

c) Label-only (reconstruction-threshold) MIA.: The autoencoder analogue of [11]. Calibrate a threshold θ_{rec} at false-

positive rate 0.05 on the auxiliary set, then count member-set rows whose reconstruction error sits below the threshold:

$$s_{\text{lo}} \;=\; \frac{1}{|M|}\sum_{x\in M} \mathbb{I}\big[\|f_{\text{post}}(x)-x\|^2 \le \theta_{\text{rec}}\big]. \tag{8}$$

3) Unlearning Attestation struct and hash binding: The Unlearning Attestation is the on-chain artefact produced by the data controller after running the three MIAs. It is a tuple

$$u \;=\; \langle i,\, h_{\text{pre}},\, h_{\text{post}},\, m,\, s^{\text{fp}}_{\text{sh}},\, s^{\text{fp}}_{\text{lira}},\, s^{\text{fp}}_{\text{lo}},\, t_u\rangle, \tag{9}$$

where i is the revoked token ID, $h_{\text{pre}}, h_{\text{post}} \in \{0,1\}^{256}$ are SHA-256 hashes of the pre- and post-unlearning model states, $m \in \{0,1,2\}$ identifies the unlearning method, $s_j^{\text{fp}} = \lfloor 10^4 \cdot s_j \rfloor$ are fixed-point versions of the three MIA scores, and t_u is the UA's Unix timestamp.

The UA hash, which every verifier signs and which the oracle re-derives to authenticate the submission, binds the chain identifier to prevent cross-chain replay:

$$h_u \;=\; \text{keccak256}\big(c \,\|\, i \,\|\, h_{\text{pre}} \,\|\, h_{\text{post}} \,\|\, m \,\|\, s^{\text{fp}}_{\text{sh}} \,\|\, s^{\text{fp}}_{\text{lira}} \,\|\, s^{\text{fp}}_{\text{lo}} \,\|\, t_u\big), \tag{10}$$

where c is `block.chainid` and $\|$ is `abi.encode`-compatible concatenation.

The acceptance predicate (Eq. 4) demands all three MIA scores below $\tau = 0.55$ (i.e., $s_j^{\text{fp}} < 5500$) and at least t distinct recovered signers from the registered verifier set VC:

$$\begin{aligned}\text{accept}(u,\Sigma) \;\Leftrightarrow\; &\big(\forall j \in \{\text{sh},\text{lira},\text{lo}\} : s_j^{\text{fp}} < 5500\big)\\ &\wedge\; \big|\{\sigma\in\Sigma : \text{recover}(\sigma, h_u) \in VC\}\big| \ge t\\ &\wedge\; h_{\text{pre}} = CR.\texttt{modelHashAtRevocation}(i)\\ &\wedge\; CR.\texttt{state}(i) \in \{\textsc{Revoked}, \textsc{Expired}\}.\end{aligned} \tag{11}$$

The third conjunct binds the UA to the specific model state captured at revocation (write-once in the registry); the fourth blocks attestation against tokens already in UNLEARNED (closing the replay loop).

C. Three-contract Solidity architecture

ConsentChain is implemented as three Solidity 0.8.24 contracts with the following responsibilities (full NatSpec interfaces in supplementary `docs/contracts-spec.md`):

a) `ConsentRegistry`: Soulbound ERC-721 contract [15]. The `_update` hook overrides `transferFrom` to revert on any non-mint transfer. Each token carries a `ConsentRecord` struct exposing the CLSM state, purpose set hash, TTL, owner address, and (after first revocation or expiry) the captured `modelHashAtRevocation`. The five DS-controlled and two time-driven δ edges are external functions. Only the registered `UnlearningOracle` address may call `transitionToUnlearned`.

b) `UnlearningOracle`: Maintains the verifier set VC as an `EnumerableSet` of n addresses, the threshold t, and the MIA threshold $\tau^{\text{fp}} = 5500$. `submitAttestation` implements Eq. 11: it re-derives h_u on-chain, recovers each submitted signature, checks membership in VC, counts distinct recoveries, validates the three MIA scores against τ^{fp}, and validates the state preconditions. On success it calls $CR.\texttt{transitionToUnlearned}(i, h_u)$ and $GK.\texttt{unblock}(i)$ in the same transaction.

c) `FLGatekeeper`: Maintains `currentRoundId`, `currentModelHash`, and the pending-unlearning bookkeeping (`pendingUnlearningCount`, `pendingUnlearningSince`). `validateRound` consults the registry's `isEligible` view for each participant and reverts if any participant is ineligible or if `pendingUnlearningCount > 0`; the latter condition is the on-chain encoding of Property 2's safety degradation. `reportTimeout` is permissionless: any caller may emit `ComplianceViolation` once block height exceeds the $\Delta_{\max}$-window deadline (Section V reports the empirical demonstration of this path).

d) Automation loop.: A revocation event at the registry fires `recordPendingUnlearning` on the gatekeeper as an internal hook. After the data controller submits a valid UA the oracle reverses this through `unblock`. The three contracts thus form a closed automation loop: state at the registry drives the gatekeeper's behaviour, and the oracle is the only legitimate path to release pending-unlearning blocks.

V. EVALUATION

We evaluate ConsentChain along four empirical axes: (i) unlearning effectiveness under three approximate methods plus an exact-retraining baseline (E6), (ii) on-chain cost on PureChain versus Polygon Amoy (E7), (iii) scalability of round validation up to $N = 500$ participants (E8), and (iv) utility recovery through the revocation event. Two short on-chain demos additionally validate Properties 2 and 3, and Section V-G cross-validates the same pipeline on CASAS. The complete reproducibility recipe lives in supplementary `docs/eval-protocol.md`.

A. Experimental setup

The primary federation is a 9-client N-BaIoT autoencoder anomaly detector [12], one client per IoT device (doorbells, thermostats, baby monitors, security cameras, webcams). Each client holds 5,000 benign-traffic rows over 115 network features; the global model is a symmetric MLP autoencoder ($115 \to 75 \to 50 \to 25 \to 10$, ReLU, dropout 0.1) trained with FedAvg over 30 rounds (Flower [16]), batch size 256, learning rate 10^{-3}, random seed 42. At round 30 we simulate a GDPR Art. 17 erasure event against client 5 (*Provision_PT_838_Security_Camera*) and trigger one of three approximate unlearning methods (SISA-Fed, gradient ascent, FedEraser) or, as the exact reference, full retraining of an 8-client federation from scratch. All experiments record an environment fingerprint (commit hash, library versions, hardware, seed) in an `env.json` alongside the metric JSON.

On-chain experiments run against the live contracts deployed on both PureChain (chainId `900520900520`, gas-free, PoA[2]) and Polygon Amoy (chainId `80002`, gas-based, PoS). The verifier set VC is five ECDSA keys with threshold $t = 3$; $\Delta_{\max} = 1000$ blocks unless otherwise noted.

TABLE III

TABLE 1 — UNLEARNING EFFECTIVENESS (E6). MIA SCORES AGAINST THE REVOKED CLIENT'S TRAINING DATA, EVALUATED ON THE POST-UNLEARNING MODEL. THE ON-CHAIN GATE ACCEPTS A UA IFF ALL THREE ARE BELOW $\tau = 0.55$.

Method	Shadow MIA	LiRA	Label-only	On-chain accept
No unlearning (baseline)	0.5455	0.0475	0.0060	✓
Gradient ascent	0.4380	0.0435	0.0030	✓
SISA-Fed	0.2542	0.0795	0.0862	✓
FedEraser	0.5232	0.0325	0.0050	✓
Naive retrain (exact ref.)	0.5843	0.0445	0.0138	×

TABLE IV

TABLE 2 — PER-OPERATION GAS AND COST (E7), LIFECYCLE ×10 MEAN. PURECHAIN COSTS ARE EXACTLY $0; AMOY COSTS ARE NON-ZERO. PER-DS LIFECYCLE = SUM OF THE FOUR OPS.

Operation	PureChain gas	Amoy gas	Amoy MATIC	Amoy USD
`updateModelHash`	30,074	32,673	0.0010	$0.00009
`grantConsent`	144,868	153,568	0.0046	$0.00044
`revokeConsent`	107,904	96,404	0.0029	$0.00028
`submitAttestation` (3-of-5)	102,903	137,600	0.0041	$0.00039
Per-DS lifecycle	**385,750**	**420,245**	**0.0126**	**$0.00120**

B. Unlearning effectiveness

Table III reports each method's three independent MIA scores against the revoked client's member set. The on-chain `submitAttestation` predicate (Eq. 11) accepts a UA iff all three scores sit below $\tau = 0.55$.

Three observations stand out. All three approximate methods produce UAs the chain accepts (shadow 0.438/0.254/0.523 for gradient ascent/SISA-Fed/FedEraser; LiRA and label-only well below τ). The naive-retrain baseline — a model that *never saw* client 5's data — yields shadow $0.584 > \tau$ and is *rejected*: shadow picks up client 5's distinctive traffic distribution, not membership, a known confound [9] that LiRA (0.045) and label-only (0.014) do not exhibit on the same model. The protocol is thus faithful to MIA evidence, not the controller's claim (Section VI). Gradient ascent's shadow (0.438) even sits *below* naive retrain's: anti-fitting moves the model further from client 5's distribution than never seeing it. The methods trace a forgetting–utility frontier — SISA-Fed strong-forgetting, gradient ascent / FedEraser preserve-utility — made explicit by the recovery curves (Section V-E). All three UAs executed end-to-end on PureChain (`transitionToUnlearned` in the same tx as `submitAttestation`, final CR.`stateOf` = UNLEARNED; representative gradient-ascent tx `0x5beb...416a`, full records in supplementary `onchain_attestation.json`).

C. Gas cost comparison

Table IV reports per-operation gas usage and the corresponding cost on each chain, taken from a lifecycle ×10 variance run (E7). USD figures use POL/USD = 0.095347 fetched from CoinGecko on the measurement day; raw gas counts and Wei costs are deterministic and chain-state-dependent.

Per-data-subject lifecycle cost is $0 on PureChain and $0.00120 on Amoy. Gas usage differs by up to 34% between the two chains (`submitAttestation`: 102,903 on PureChain versus 137,600 on Amoy) due to opcode-pricing differences between the PoA[2] schedule and Polygon's EIP-2929-equipped schedule; the dominant cost factor at scale is the gas price (zero versus ∼30 gwei), not the gas count. Section V-D quantifies the recurring per-FL-round cost as deployments grow.

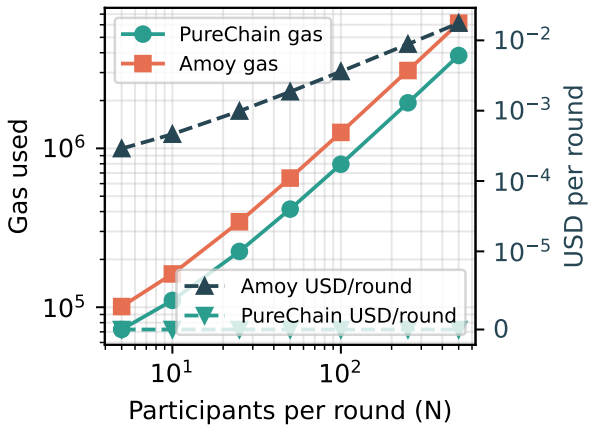


Fig. 3. Figure 1 — Scalability of `validateRound`(N) (E8). Solid lines: live gas measurements. Dashed lines: per-round USD cost. PureChain cost line is at zero across all N; Amoy projects to $0.0176 at $N = 500$.

D. Scalability of round validation

Figure 3 reports gas usage and equivalent USD cost for `validateRound`(N) as N scales from 5 to 500 participants. On PureChain the full sweep ran live; on Amoy only $N \in \{5, 10\}$ were measured directly (wallet-budget constraint) and the remaining points are extrapolated via least-squares linear fit $\text{gas}(N) \approx 40{,}002 + 12{,}202N$, which the PureChain measurements (slope 7,616) confirm as exactly linear ($R^2 \approx 1$).

PureChain's per-participant slope is ∼ 38% lower than Amoy's — the EIP-2929 cold-SLOAD pricing (2,100 gas) dominates the linear term on Amoy. The cost story is dominated by gas *price*: on Amoy a single $N = 500$ round-validation costs $0.0176; on PureChain the same operation costs $0. At a typical 100-FL-rounds-per-day cadence this is $1.76/day per deployment on Amoy versus $0 on PureChain — gas-free infrastructure is a different cost class, not an optimisation.

E. Utility recovery after revocation

Figure 4 plots the size-weighted central reconstruction error for five trajectories diverging at round 30, the revocation point. The baseline 9-client federation continues to round 40 with all clients; the other four trajectories drop client 5 and resume FL for ten further rounds, each starting from a different round-30 checkpoint.

The post-revoke trajectory ends at 0.4720, 6.1% *better* than the 9-client baseline (0.5025) — client 5 was an outlier whose removal lets the federation fit the remaining devices more tightly. Gradient ascent (0.4726) and FedEraser (0.4698) match this with no utility cost. SISA-Fed pays a real cost (0.7009 at round 31, recovering to 0.6081 by round 40 — +29%),

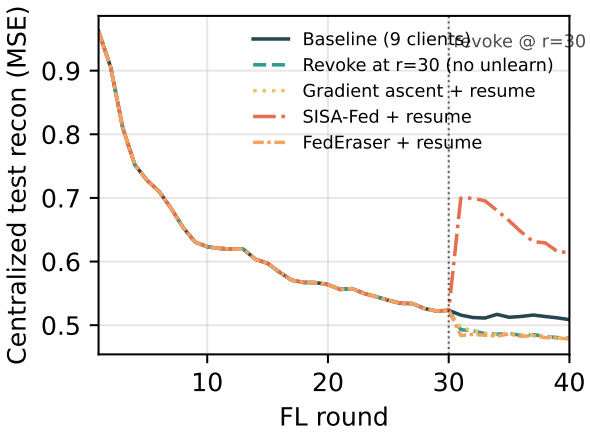


Fig. 4. Figure 2 — Utility recovery (E6). Centralized test reconstruction error per FL round. The four post-revocation trajectories diverge at round 30; SISA-Fed pays a real utility cost, gradient ascent and FedEraser converge to within 0.5% of plain post-revoke by round 40.

consistent with its strongest-forgetting position and lowest shadow MIA in Table III.

F. On-chain property demos

Two short PureChain runs validate the formal properties. **Liveness** (Property 2): with $\Delta_{\max} = 2$, `reportTimeout` reverts (`NotYetTimedOut`) pre-deadline and emits `ComplianceViolation` (24,800 gas) once past it. **Verifiability** (Property 3): a t-of-5 sweep over $t \in \{2, 3, 4, 5\}$ confirms both branches of Eq. 4 — t signatures accept, $t-1$ revert with `InsufficientSignatures` (marginal $\sim$7,800 gas/sig; default $t=3$ totals 102,941 gas).

G. Cross-domain validation (CASAS)

We replayed Section V-B–V-E on the CASAS smart-home benchmark (5-client BiLSTM, per-apartment sensor streams, `twor_2010` revoked at round 60). The distribution-shift confound is sharper here: shadow rejects pre-unlearn, naive retrain, gradient ascent, and FedEraser (all ≥ 0.84), so SISA-Fed ($0.499/0.035/0.000$ shadow/LiRA/label) is the *only* method passing the $\tau = 0.55$ gate — the strongest-forgetting Pareto endpoint, consistent with N-BaIoT. A 15%-ratio backdoor canary triggers 100% pre-unlearning and drops to 0% under SISA-Fed and FedEraser (gradient ascent preserves it at 10 steps). End-to-end SISA-Fed attestation on PureChain (`submitAttestation 0x9643...733b`) transitioned token 527 to UNLEARNED.

H. Comparison to prior work

Table V positions ConsentChain against the three closest prior systems; each cell states the system's actual mechanism, so the per-dimension comparison is read directly off the table. The takeaway: MedRec, FedEraser, and BlockFL each cover one dimension (consent, unlearning, on-chain logging respectively) but none gates training on verified unlearning. ConsentChain is the first design we are aware of to address all four dimensions in a single chain-enforced protocol, and the first to do so at gas-free per-operation cost on a public EVM-compatible substrate.

VI. DISCUSSION

Limitations. Shadow MIA conflates a client's distributional uniqueness with membership — naive retrain scored 0.584/0.846 (N-BaIoT/CASAS) despite never seeing the data; LiRA resists this but is α-sensitive, so the gate is sound only because it demands *all three* MIAs below τ and deployments must co-calibrate (α, τ) per model class. ConsentChain accepts *approximate* unlearning (exact retraining is infeasible at IoT scale; GDPR Art. 17's "reasonable measures" is met by Eq. 4 with the chain as audit evidence), leaving formal-erasure guarantees and ECDSA verifier-identity leakage (BLS-collapsed in the journal version) out of scope. Off-chain training on revoked data is unpreventable but leaves no UA binding the deployed model hash to a valid consent state — a publicly detectable omission.

Regulatory & deployment. ConsentChain discharges GDPR Arts. 7/17/21, EU AI Act Arts. 15/26, and CCPA § 1798.105 via the soulbound NFT, the $\Delta_{\max}$-bounded liveness and verifiable-unlearning predicates, and audit completeness (Properties 1–4; article-by-article mapping in supplementary `docs/regulatory-alignment.md`). PureChain's PoA^2 is validator-permissioned but publicly readable and submittable (apt for known-stakeholder IoT consortia); the Amoy deployment shows the identical contracts on a permissionless PoS chain at non-zero cost — the economic model is the decision variable, not the protocol semantics.

Future work: (i) a zk-SNARK over the CLSM transition predicate so `isEligible` proves eligibility without leaking the subject's identity or purposes; (ii) PoA^2-native verifiers re-running the MIAs at block production (consensus-time UA acceptance); (iii) cross-chain consent portability via a light-client bridge.

VII. CONCLUSION

We presented ConsentChain, the first protocol we are aware of to bind dynamic consent, machine unlearning, and on-chain verification into a single chain-enforced pipeline — closing the consent–unlearning–verification triangle that prior work addressed only at the vertices. The protocol decomposes into a formally-stated Consent Lifecycle State Machine implemented as soulbound ERC-721 tokens, a method-agnostic Verifiable Unlearning Protocol whose acceptance predicate is the conjunction of three independent MIA scores and a t-of-n ECDSA committee signature, and a three-contract Solidity architecture that runs unchanged on gas-free and gas-based EVM substrates. End-to-end empirical evaluation on a 9-client N-BaIoT [12] federated autoencoder, cross-validated on a 5-client CASAS BiLSTM activity classifier, confirms the four headline numbers: per-data-subject on-chain lifecycle cost of \$0 on PureChain versus \$0.00120 on Polygon Amoy, `validateRound` scaling linearly to $N = 500$ at \$0 on PureChain versus a projected \$0.0176 on Amoy, MIA scores

TABLE V

TABLE 3 — FEATURE COMPARISON WITH THE THREE CLOSEST PRIOR SYSTEMS. EACH CELL DESCRIBES THE CORRESPONDING SYSTEM'S MECHANISM FOR THAT DIMENSION; "NOT PROVIDED" MARKS DIMENSIONS THE PRIOR SYSTEM WAS NOT DESIGNED TO ADDRESS, NOT FAILURES OF IMPLEMENTATION. PER-OPERATION COST IS OMITTED HERE — SEE TABLE IV.

Dimension	ConsentChain (ours)	MedRec [4]	FedEraser [6]	BlockFL [7]
Consent representation	6-state CLSM on a soulbound ERC-721 token; every transition signed by the data subject's wallet	Per-record access policy on a permission contract, owned by the patient	Not provided — consent is managed off-chain by the deployer	Not provided — participation is implicit on join
Unlearning trigger & method	Chain-gated trigger ($\Delta_{\max}$-bounded liveness, Property 2); method-agnostic — accepts any (pre, post) model-hash pair from SISA-Fed, gradient ascent, FedEraser, or future methods	Not applicable: no downstream ML training	Trigger left to the deployer; method fixed to FedEraser's history-replay-with-calibration scheme	Not applicable: no unlearning step
Verification of unlearning	Three independent MIAs (shadow, LiRA, label-only) all below τ *and* t-of-n ECDSA signatures from a registered verifier committee (Property 3)	Not applicable	Self-reported by the data controller; no external check	Not applicable
On-chain footprint & audit trail	3 Solidity contracts (registry, oracle, gatekeeper) on either gas-free PureChain or gas-based Polygon Amoy; every CLSM transition, UA acceptance, and `ComplianceViolation` emitted as an indexed event	Permission contracts on Ethereum mainnet; public ledger of permission grants/revocations	None — entirely off-chain; no immutable record	Per-round device-side model-update log on Ethereum mainnet; public ledger of update commitments

driven below the on-chain $\tau = 0.55$ threshold by approximate unlearning (all three methods on N-BaIoT; SISA-Fed on CASAS), and an absolute utility delta of $\leq 3\%$ on two of three methods — in fact a 6.1% utility *gain* when the revoked client was an outlier in the benign distribution. Future work targets BLS threshold-signature aggregation, zero-knowledge consent eligibility proofs, PoA^2-native verifier-committee integration, and cross-chain consent portability for federations spanning multiple EVM chains.

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